Number Systems

The first numbers one encounters are the *natural numbers*, which are the counting numbers

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}.$$

We can perform addition perfectly well in $\mathbb{N}$, but if we want to subtract we must extend our number system to the *integers*, which also contains 0 and the negative of each natural number:

$$\mathbb{Z} =$$

The next issue is multiplication and division. We can multiply perfectly well in $\mathbb{Z}$ (and in $\mathbb{N}$), but in order to define division we must extend our number system again to the *rational numbers*

$$\mathbb{Q} =$$

Unlike the number line for $\mathbb{Z}$, the number line for $\mathbb{Q}$ has no intervals of empty space:

Proposition: Given any two rational numbers $r < s$, there is another rational number between them.

Despite this, the number line for $\mathbb{Q}$ still has "holes"; that is, there are lengths on the number line that do not correspond to any rational number. For example, $\sqrt{2}$, which is the length of the diagonal of the unit square, is not rational.
Proposition. There is no rational number x such that $x^2 = 2$.

We will prove this proposition using a technique called *proof by contradiction*, which proceeds as follows:

Before we begin, we need a preliminary result about even and odd numbers. First, recall

Fundamental Theorem of Arithmetic. Any natural number n can be written uniquely as $n = p_1 p_2 \cdots p_k$, where each p_i is prime.

For example:

Lemma. A natural number n is even if and only if n^2 is even.

Proof of lemma.

Proof of proposition.

The Real Number System

Since there is no rational number x such that $x^2 = 2$, there is a hole in the rational number line where $\sqrt{2}$ ought to be. There are many other holes as well. For example, one can similarly prove that if n is an integer which is not a perfect square, then there is no rational number x such that $x^2 = n$. If we want all of these lengths on the number line to correspond to an actual number, then we need to extend our number system again. Thus, to the chain $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$ we append the *real numbers* $\mathbb{R}$:

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}.$$

The question of how to actually construct $\mathbb{R}$ from $\mathbb{Q}$ is rather complicated, and is beyond the scope of this course. For now, it is not too inaccurate to say that $\mathbb{R}$ is obtained by filling in the holes in $\mathbb{Q}$. Wherever there is a hole, a new *irrational* number is defined an placed into the ordering that already exists on $\mathbb{Q}$. The real numbers are then the union of these irrational numbers together with $\mathbb{Q}$, and the operations of $\mathbb{Q}$ extend to all of $\mathbb{R}$ in such a way that we can perform all of the standard manipulations of addition, subtraction, multiplication, and division that are second nature to us.

For example, all of the following manipulations are valid in $\mathbb{R}$:

Exercises

1. (a) Prove that a natural number n is divisible by 3 if and only if n^2 is divisible by 3.
   (b) Prove that there is no rational number x such that $x^2 = 3$.
2. Prove that if p is prime, then there is no rational number x such that $x^2 = p$.